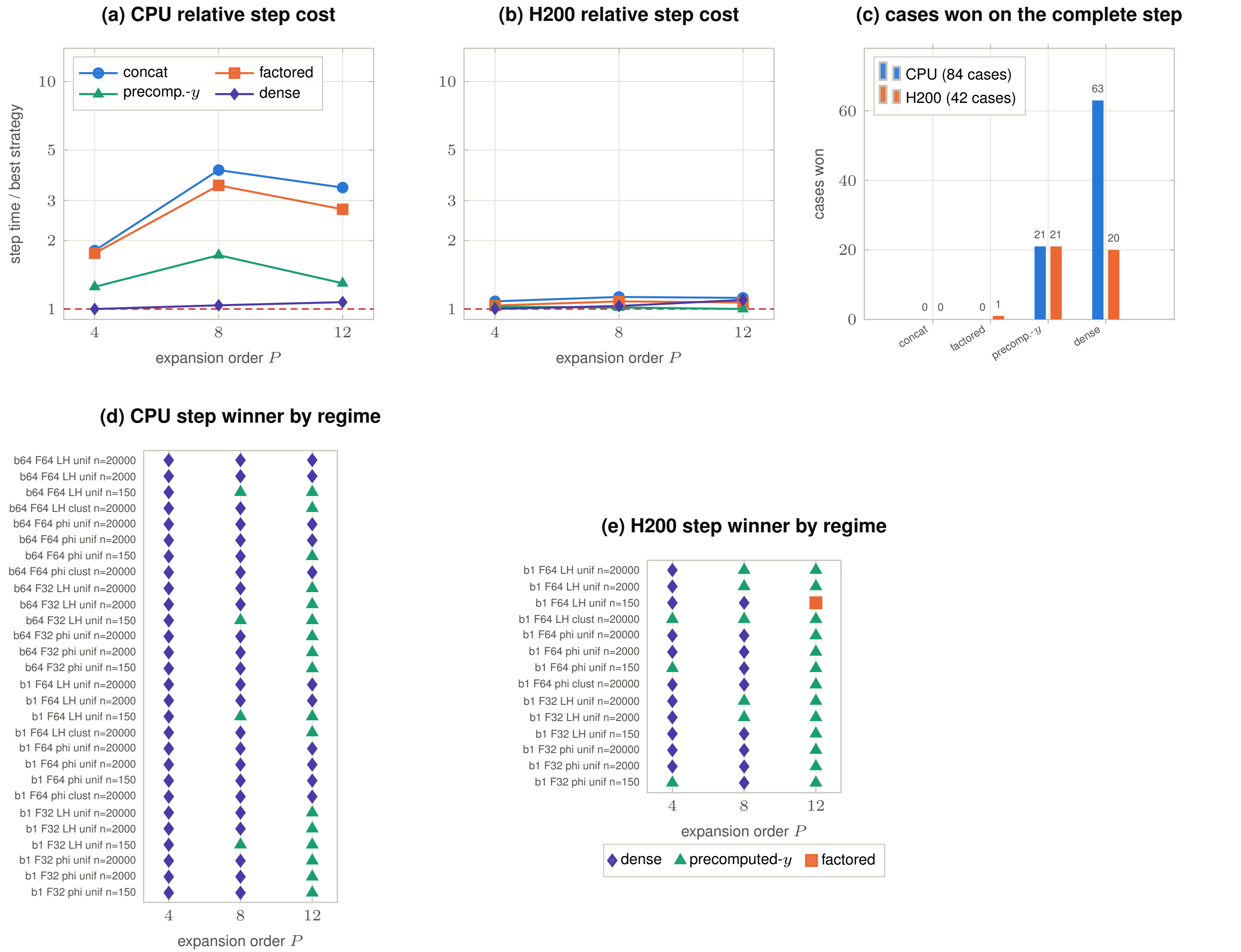
Fig. 5 — Which of the four resident M2L strategies to use, by platform and regime (task 024)



Machines / regimes: CPU = `m12-2-17`, AMD EPYC 7763, Slurm job 12892009, two processes with OpenBLAS pinned before startup to 1 and 64 threads (`b1`/`b64` in the regime labels), `julia_threads=1` throughout. GPU = `m13h-1-1`, NVIDIA H200, Slurm job 12892013, device-resident lifecycle. 126 cases $\times$ 4 strategies; 483 eligible rows, 21 infeasible, 0 disqualified; every eligible row passed its correctness gate. 2 warmups, 7 samples, medians. **Data:** `operator_ab_benchmark/summary/{case_rankings,cpu_crossovers,gpu_crossovers}.csv` (tables `fig05a_*`, `fig05b_*`, `fig05{cpu,gpu}_win_*`). **Reads as:** on the CPU the choice is binary — dense takes 63 of 84 cases and precomputed-y the other 21; `concat` and `factored` win nothing and sit 1.8–4.1$\times$ off the best. On the H200 the split is nearly even (20 dense / 21 precomputed-y / 1 factored), and the pattern is orderly: dense owns $P = 4$ and φ-only $P = 8$, precomputed-y owns $P = 12$ and Lamb–Helmholtz at larger n. Neither n nor mean class occupancy alone predicts the winner — channel and order reverse it at matched occupancy. **Caveats:** panels (a,b) are geometric means over all cases at that P, so they compress regimes that (d,e) separate; strategies infeasible in a case are omitted from that case's mean rather than penalised. "Multi-thread CPU" here means BLAS threads only. **Scope:** the ranked quantity is the complete recurring `fmm!` step — state refresh, the full B2M→M2M→M2L→L2L→L2B lifecycle including the *nearfield* direct pairs evaluated inside L2B, and output writeback — so these are end-to-end evaluations, gated for every case against an all-bodies `direct!` reference. The M2L-only ranking (which flips 6 of 42 H200 winners) is the kernel subset of the same runs. Three eligible CPU rows had M2L IQR/median above 10% (0.113, 0.195, 0.713) and were recorded as unresolved noise by task 024. The reconstructed per-column $T_s(\theta)$ path is a correctness oracle and is excluded from ranking. Task 024 changed no production defaults — these are recommendations, not shipped dispatch.